LPASU-SwinDet: A Multiscale Transformer Framework for Maritime Infrared Small Target Detection

Dongdong Ma, *Member, IEEE*, Tingkuo Chen, and Wenhai Xu

***Abstract*—To address the challenges of severe background clutter interference, low signal-to-clutter ratio, and weak long-range semantic dependence modeling in maritime infrared small target detection, this letter proposes lightweight pose-aware subpixel upsampling (LPASU)-SwinDet, a multiscale Transformer-based detection framework. Built on a modified YOLO26 backbone, the network introduces the LPASU module to replace conventional nearest-neighbor upsampling, enhancing detail reconstruction in feature fusion while suppressing background clutter. To further capture long-range contextual information and distinguish small targets from sea clutter, Swin Transformer-based detection heads are employed to improve modeling of long-range semantic dependencies, boosting detection robustness and localization accuracy in complex marine scenes. In addition, the normalized Wasserstein distance (NWD) loss and Wasserstein distance loss (WDL) are introduced to stabilize gradients for tiny targets and alleviate the positive–negative sample imbalance problem in bounding box regression. Extensive experiments on maritime infrared datasets demonstrate that the proposed method achieves superior detection accuracy and strong robustness against various challenging sea conditions.**

***Index Terms*—Feature fusion, maritime infrared target detection, Swin transformer.**

I. Introduction

INFRARED detection of maritime distress targets is essential for infrared search and tracking systems and marine emergency rescue. However, long-range targets usually occupy only a few pixels and are easily obscured by sea clutter, resulting in low signal-to-clutter ratios and frequent missed or false detections. Therefore, effective target enhancement and clutter suppression remain key challenges in maritime infrared small-target detection [1].

Existing algorithms can be divided into traditional model-driven methods and deep learning-based detection methods. Model-driven methods include estimation filtering [2], local contrast measures [3], and sparse low-rank representation [4]. Although effective in specific scenarios, these methods depend heavily on handcrafted features and predefined assumptions, limiting their robustness in complex environments. With the development of deep learning, infrared small-target detection methods can be broadly divided into three categories.

At present, deep learning-based detection methods are prevailing, including Faster R-CNN [5], the YOLO series [6], [7], [8], and Transformer-based RT-DETR [9]. The second category improves existing detectors using attention mechanisms, feature-enhancement modules, or lightweight structures [10]. For example, Xin et al. [11] introduced attention modules into YOLOv5s, Guo et al. [12] combined an FC3 backbone with Swin Transformer, Zhang et al. [13] developed a lightweight YOLOv9 detector, and Li et al. [14] proposed MFE-Net for enhanced faint-target representation. The third category designs task-specific infrared detection networks. Representative methods include learnable saliency kernels [15], wavelet- and Mamba-based feature modeling [16], causal-guided target verification [17], nonuniformity correction [18], noise-suppression feature pyramids [19], and multimodal infrared prior modeling [20].

Despite these advances, existing methods still struggle to balance feature preservation, clutter suppression, localization accuracy, and computational efficiency in complex maritime scenes. General-purpose detectors are insufficiently adapted to tiny targets and strong sea clutter, while many improved models rely on generic modules that cannot fully preserve weak target details during multiscale fusion. To address these limitations, we propose lightweight pose-aware subpixel upsampling (LPASU)-SwinDet, which integrates lightweight subpixel upsampling, Swin Transformer detection heads, and Wasserstein-distance-based bounding-box regression. The main contributions are given as follows.

1) We replace the original nearest-neighbor upsampling with the LPASU module to enhance detail reconstruction in feature fusion and suppress background clutter interference.
2) We employ the Swin Transformer as the detection head to enhance long-range semantic dependence modeling, which effectively distinguishes infrared small targets from sea clutter and further improves detection robustness and positioning accuracy in complex marine scenes.

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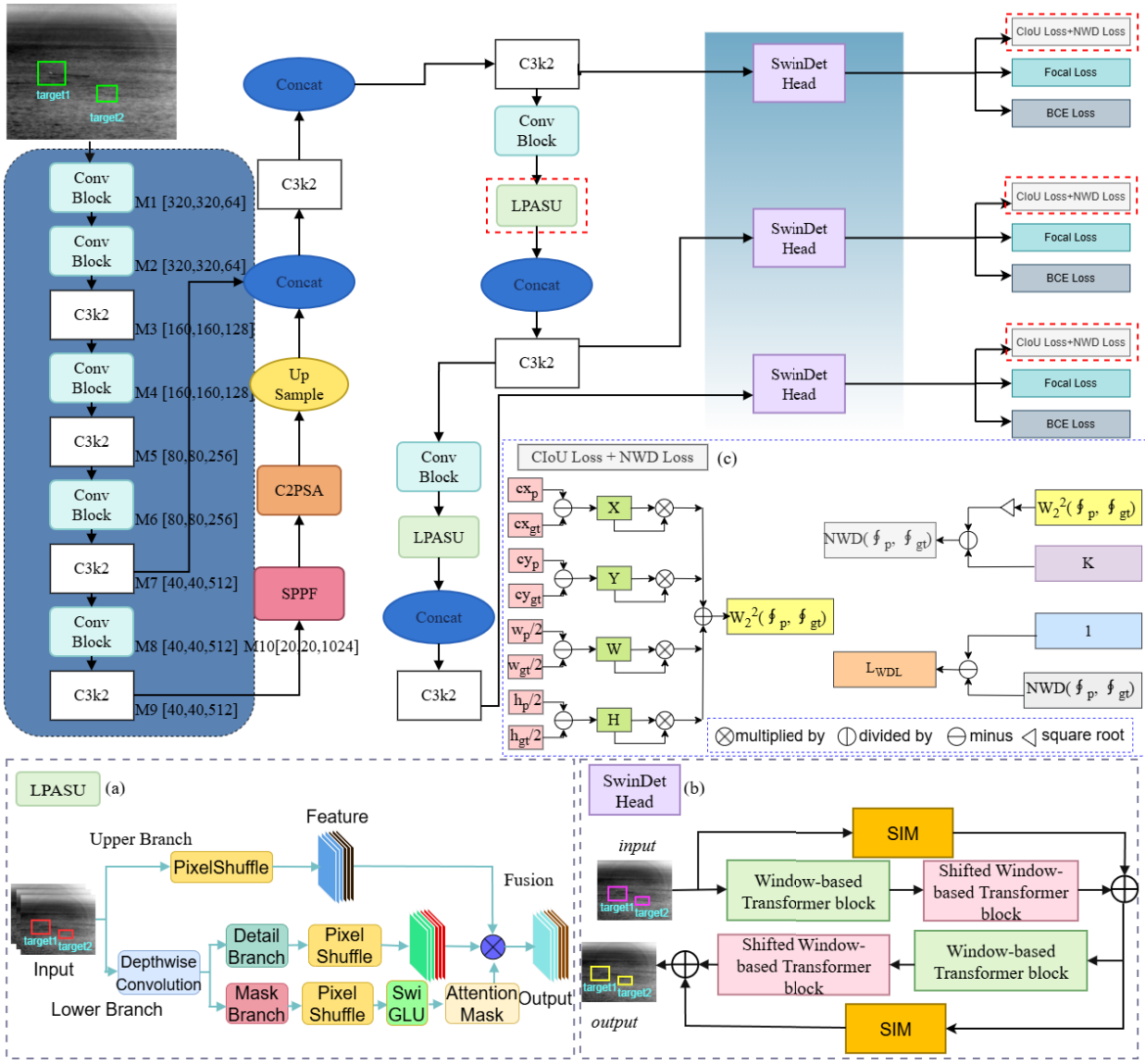


Fig. 1. Architecture of the proposed method. (a) LPASU module. (b) SwinDet Head. (c) NWD loss.

3) We introduce the Wasserstein distance loss (WDL) function based on the Wasserstein distance to replace CIoU loss, which stabilizes gradients for tiny targets and alleviates the problem of positive–negative sample imbalance in bounding box regression.

II. Methodology

A. Overview of LPASU-SwinDet

The overall architecture of LPASU-SwinDet is a multiscale Transformer-based framework for maritime infrared small target detection built on a modified YOLO26 backbone, as shown in Fig. 1. The figure details three core components: (a) LPASU module, which replaces conventional nearest-neighbor upsampling to enhance detail reconstruction during feature fusion and suppress background clutter; (b) the SwinDet Head, a Swin Transformer-based detection head that captures long-range contextual information to distinguish small targets from sea clutter and improve localization accuracy; and (c) the normalized WDL, which stabilizes gradients for tiny targets and mitigates positive–negative sample imbalance in bounding box regression, working in tandem with CIoU, Focal, and BCE losses.

B. LPASU Module

In infrared sea-surface small target detection, upsampling in the feature pyramid dominates shallow high-resolution feature reconstruction. The original nearest-neighbor upsampling causes blurred edges and lost texture, especially for low-contrast small targets. To mitigate this, we propose LPASU to replace fixed upsampling layers, enhancing detail recovery while suppressing background noise. Given input feature $X \in \mathbb{R}^{C\times H\times W}$ with upsampling factor $s = 2$, we first compress channels via a squeeze module

$$C_{\text{mid}} = \max\left(\frac{C}{4}, 16\right). \tag{1}$$

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This module uses 11 convolutions, batch normalization, SiLU, and grouped 33 depthwise separable convolutions to obtain lightweight intermediate feature $M \in \mathbb{R}^{C_{\text{mid}} \times H \times W}$. The intermediate feature is split into two parallel branches. The detail enhancement branch expands channels to $C \times s^2$ and uses subpixel convolution to output $D \in \mathbb{R}^{C \times sH \times sW}$, avoiding checkerboard artifacts and reconstructing high-frequency details. The adaptive mask branch compresses channels to $1 \times s^2$, applies subpixel convolution, and uses sigmoid to generate a spatial attention mask Mask $\in \mathbb{R}^{1 \times sH \times sW}$ within $[0, 1]$. A bilinear upsampling path is retained to produce basic feature $B \in \mathbb{R}^{C \times sH \times sW}$. The final output is

$$\text{Output} = B \odot (1 - \text{Mask}) + D \odot \text{Mask} \tag{2}$$

where $\odot$ denotes elementwise multiplication. The mask dynamically weights detail features at target edges and smooth interpolation in flat backgrounds. LPASU is lightweight, direction-aware for accurate bounding-box regression, and plug-and-play to replace any two upsampling layers.

C. Bounding Box Regression Loss Based on Wasserstein Distance

In object detection, the bounding box regression loss directly affects the localization accuracy of detected targets. YOLOv5 originally adopts the CIoU loss, which considers the overlap area, center-point distance, and aspect ratio between the predicted box and the ground-truth box. While CIoU performs well in general object detection, it exhibits two significant deficiencies in the context of infrared maritime small-target detection. First, CIoU is extremely sensitive to small positional shifts of small targets. For targets with dimensions smaller than 16×16 pixels, a tiny pixel offset between the predicted box and the ground-truth box can cause the IoU value to drop sharply from above 0.5 to below 0.1, resulting in severe fluctuations in the loss function, unstable gradients, and difficulty in model convergence. Second, there is a severe imbalance between positive and negative samples. The sea-surface background accounts for more than 95% of the image area; the IoU loss of a large number of negative samples is close to zero, whereas positive samples are scarce and have much larger loss values. Consequently, the model training tends to bias toward background suppression, and the learning of small and weak targets is insufficient. The traditional CIoU loss cannot effectively address this issue.

To tackle the above problems, this letter introduces a bounding box regression loss function based on the Wasserstein distance, termed WDL. The Wasserstein distance measures the discrepancy between two probability distributions. Its key advantage is that even when the overlap between the two distributions is very small or even zero, the Wasserstein distance still provides smooth and meaningful gradient information. For small-target detection, after modeling a bounding box as a 2-D Gaussian distribution, the Wasserstein distance effectively alleviates the sensitivity of IoU-based losses to small displacements and simultaneously balances the gradient contributions of positive and negative samples.

Specifically, each horizontal bounding box $B(c_x, c_y, w, h)$ is modeled as a 2-D Gaussian distribution $\Phi_{\mu,\Sigma}$, where the center coordinates form the mean vector $\boldsymbol{\mu} = (c_x, c_y)$ and the covariance matrix is set to a diagonal matrix $\boldsymbol{\Sigma} = \text{diag}((w/2)^2, (h/2)^2)$. The physical interpretation of this modeling is that pixels near the target center have a higher probability of belonging to the target, while edge pixels have a lower probability, which is consistent with the gray-scale distribution characteristics of infrared small and weak targets (bright center and blurred edges). For a predicted box B_p and the ground-truth box B_{gt}, their corresponding Gaussian distributions are denoted as $\Phi_p(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ and $\Phi_{gt}(\boldsymbol{\mu}_{gt}, \boldsymbol{\Sigma}_{gt})$, respectively.

For diagonal covariance matrices, the Wasserstein distance between the two distributions is defined as

$$W_2^2\left(\Phi_p, \Phi_{gt}\right) = \left(c_{x_p} - c_{x_{gt}}\right)^2 + \left(c_{y_p} - c_{y_{gt}}\right)^2 + \left(\sqrt{\frac{w_p}{2}} - \sqrt{\frac{w_{gt}}{2}}\right)^2 + \left(\sqrt{\frac{h_p}{2}} - \sqrt{\frac{h_{gt}}{2}}\right)^2. \tag{3}$$

To further normalize the Wasserstein distance to the interval $[0, 1]$ and use it as a loss function, this letter adopts the normalized Wasserstein distance (NWD), expressed as

$$\text{NWD}\left(\Phi_p, \Phi_{gt}\right) = \exp\left(-\frac{\sqrt{W_2^2\left(\Phi_p, \Phi_{gt}\right)}}{K}\right) \tag{4}$$

where K is a hyperparameter controlling the decay speed of the distance. Based on the statistical distribution of target scales in our dataset, we set $K = 5$. Finally, the WDL function is defined as

$$\mathcal{L}_{\text{WDL}} = 1 - \text{NWD}\left(\Phi_p, \Phi_{gt}\right). \tag{5}$$

The range of this loss function is $[0, 1]$. When the predicted box exactly coincides with the ground-truth box, $W_2^2 = 0$, NWD $= 1$, and the loss is zero; as the deviation increases, the loss increases smoothly.

D. Swin Transformer Detection Head

To improve the discrimination of weak targets from sea clutter, we introduce the NWD loss and WDL, which is weight fused with the CIoU loss to stabilize gradients for tiny targets and mitigate the scarcity of positive anchors for maritime infrared small targets. As shown in Fig. 1(b), the SwinDet Head alternates between window-based multihead self-attention (W-MSA) and shifted-window MSA (SW-MSA). W-MSA captures local target details within nonoverlapping windows, while SW-MSA enables cross-window information exchange and expands the contextual receptive field.

Given an input feature map $\mathbf{X}^{l-1} \in \mathbb{R}^{H \times W \times C}$, two consecutive Swin Transformer blocks are formulated as

$$\begin{aligned}
\hat{\mathbf{X}}^{l} &= \mathbf{X}^{l-1} + \text{W-MSA}\left(\text{LN}\left(\mathbf{X}^{l-1}\right)\right) \\
\mathbf{X}^{l} &= \hat{\mathbf{X}}^{l} + \text{MLP}\left(\text{LN}\left(\hat{\mathbf{X}}^{l}\right)\right) \\
\hat{\mathbf{X}}^{l+1} &= \mathbf{X}^{l} + \text{SW-MSA}\left(\text{LN}\left(\mathbf{X}^{l}\right)\right) \\
\mathbf{X}^{l+1} &= \hat{\mathbf{X}}^{l+1} + \text{MLP}\left(\text{LN}\left(\hat{\mathbf{X}}^{l+1}\right)\right).
\end{aligned} \tag{6}$$

The attention operation in each window is computed as

$$\text{Attention}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}} + \mathbf{B}\right)\mathbf{V} \tag{7}$$

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where **Q**, **K**, and **V** denote the query, key, and value matrices, d_k is the key dimension, and **B** is the learnable relative position bias.

Three SwinDet Heads are applied to the fused features at different scales, followed by detection branches for classification and bounding-box regression. By jointly modeling local target details and cross-window context, the proposed head suppresses sea-clutter interference and improves the robustness and localization accuracy of maritime target detection.

III. Experiment

Beyond the main results presented in this section, supplementary results, including computational efficiency analysis, full quantitative comparison with state-of-the-art methods, three groups of ablation studies, sensitivity analysis of the normalization parameter, and scalewise precision–recall (PR) curve analysis, are provided in supplementary material S1–S6.

A. Experimental Settings

1) Infrared Marine Datasets: Our self-built maritime infrared dataset contains 112 video sequences with 2486 infrared frames and 5729 annotated maritime small target instances. All images have a fixed resolution of 640 × 512 pixels. Target sizes range from 2 × 2 to 32 × 32 pixels; over 86% of targets are smaller than 16 × 16 pixels. Only one target class is annotated. To prevent information leakage from highly correlated adjacent frames, we adopt sequence level instead of frame-random splitting. The sequences are split into 72 for training, 20 for validation, and 20 for testing, yielding an approximate 70%/15%/15% frame ratio. No video sequences overlap across the three subsets.

2) Evaluation Metrics: To comprehensively evaluate the performance of the proposed maritime infrared small target detection method, we adopt four quantitative metrics shown in the table header: *precision*, *recall*, mAP_{50}, and mAP_{50-95}, along with the PR curve for visual evaluation. These metrics assess the model's precision, recall, and localization accuracy under different IoU thresholds, while the PR curve further reflects the overall detection capability across varying confidence levels.

B. Comparison to the State-of-the-Art Methods

To verify the effectiveness of the proposed method, we compare it with several representative approaches, including RLCM, PSTNN, AGPCNet, Faster RCNN, YOLOv8s, YOLOv11s, YOLOv26, and RTDETR. All compared methods follow their original published configurations. For LPASU-SwinDet, the AdamW optimizer with an initial learning rate of 1e−4 and cosine annealing learning rate decay is used. We set the batch size to 8 and the total training epochs to 150. The confidence threshold is 0.25, and the NMS IoU threshold is 0.6. All experiments run on a single NVIDIA RTX 3090 GPU. For the SwinDet detection head, the window size is 8; each head stacks two Swin Transformer blocks with four multihead attention heads and an embedding dimension of 128. Computational efficiency analysis is given in supplementary material S1.

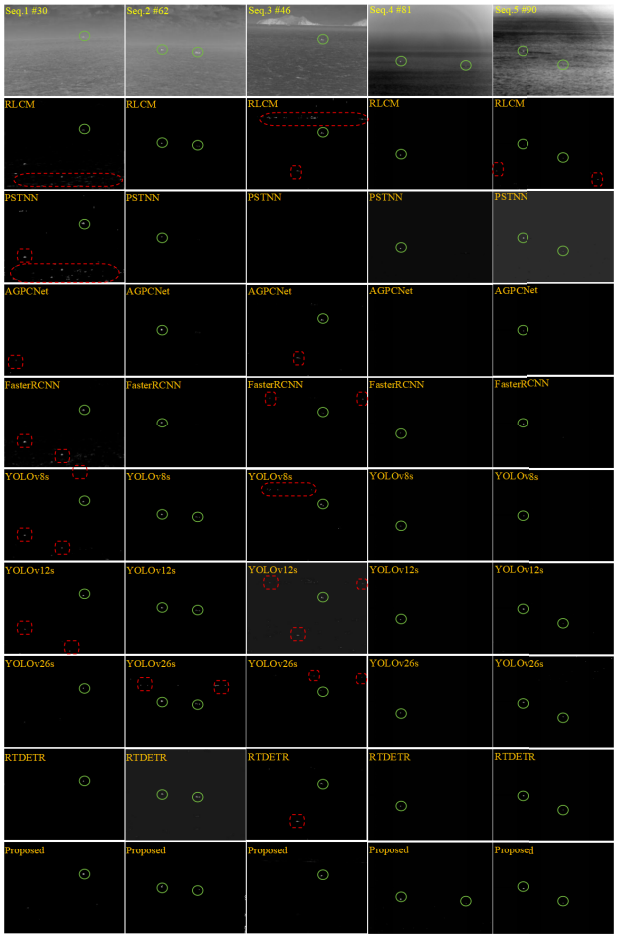


Fig. 2. Comparison of target size distribution across different datasets.

1) Ablation Studies: We conduct three sets of ablation experiments to verify the contributions of each component: core module validation, internal branch analysis of the LPASU module, and loss function configuration exploration. Due to page limits, complete ablation results and analysis are presented in supplementary material S2–S4.

2) Qualitative Comparisons: To visually demonstrate the superiority of the proposed method in complex maritime infrared scenes, we compare its detection results with those of competing methods on representative sequences, as shown in Fig. 2. The green circles mark correctly detected targets, while red dashed boxes indicate false positives or missed detections. RLCM, PSTNN, and AGPCNet are prone to false alarms caused by sea clutter and strong background noise (e.g., Seq. 1 and Seq. 3), failing to consistently suppress background interference. Meanwhile, mainstream deep learning detectors, including Faster R-CNN, YOLOv8s, YOLOv11s, YOLOv26s, and RT-DETR, exhibit varying degrees of missed detections or false positives when facing extremely small targets, low signal-to-clutter ratios, and complex backgrounds (e.g., Seq. 2 and Seq. 4). In contrast, the proposed method accurately detects all true targets without introducing false positives, demonstrating its strong anti-interference capability and robustness in distinguishing small infrared targets from complex sea backgrounds.

3) Quantitative Comparisons: The quantitative comparison results are summarized in Table I, where the proposed method

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TABLE I
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TEST SET

Method	Precision	Recall	F_1	AP_{50}	AP_{75}	mAP_{50-95}
RLCM	0.612	0.487	0.542	0.423	0.187	0.201
PSTNN	0.658	0.523	0.583	0.468	0.215	0.234
AGPCNet	0.703	0.581	0.636	0.527	0.268	0.289
Faster R-CNN	0.756	0.672	0.711	0.634	0.358	0.372
YOLOv8s	0.801	0.734	0.766	0.728	0.445	0.451
YOLOv11s	0.823	0.768	0.795	0.776	0.492	0.498
YOLOv26s	0.847	0.795	0.820	0.861	0.543	0.522
RT-DETR	0.815	0.751	0.781	0.749	0.467	0.473
LPASU-SwinDet (Ours)	**0.886**	**0.842**	**0.863**	**0.892**	**0.601**	**0.568**

achieves the optimal performance across all evaluation metrics, including precision, recall, F_1, AP_{50}, AP_{75}, and mAP_{50-95}. Compared with traditional methods (RLCM, PSTNN, and AGPCNet) and mainstream deep learning detectors (Faster R-CNN, YOLOv8s, YOLOv11s, YOLOv26s, and RT-DETR), our method yields consistent improvements on all metrics, verifying its superior detection capability and localization accuracy for maritime infrared small targets. The sensitivity analysis of the normalization parameter K in WDL and the PR curve analysis broken down by target scale are provided in supplementary material S5 and S6, respectively.

IV. CONCLUSION

In this work, we present LPASU-SwinDet for maritime infrared small target detection to address background clutter interference and poor modeling of long-range semantic information. Equipped with the LPASU upsampling module, SwinDet Head, and NWD loss, the proposed framework enhances feature detail reconstruction, long-range dependence modeling, and bounding box regression optimization. Experimental results, including qualitative and quantitative comparisons, show that our method achieves higher precision, recall, and mAP values than existing mainstream approaches. It can effectively detect tiny infrared targets under low signal-to-clutter ratio and complex sea backgrounds, and possesses excellent anti-interference ability.

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